

Notebook 1 – Exploratory Data Analysis and Feature Selection

Purpose

This notebook performs the initial preparation and exploratory analysis of the 2016 Toronto Neighbourhood Profile dataset.

The objectives of this notebook are to:

- import and inspect the raw dataset;
- evaluate data quality and completeness;
- identify candidate socioeconomic indicators;
- define the dependent (target) and independent (predictor) variables;
- perform feature selection and justification;
- generate a machine-learning-ready dataset for subsequent analyses.

This notebook addresses the first stage of the predictive analytics workflow and provides the foundation for the relationship analysis, dimensionality reduction, clustering, and predictive modelling notebooks.

```
In [2]: import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from ydata_profiling import ProfileReport

pd.set_option("display.max_columns", None)

print("Libraries imported successfully.")
```

Libraries imported successfully.

C:\Users\local_ksalyamova\Temp\ipykernel_15976\3755764946.py:7: DeprecationWarning:
`import ydata\_profiling` is deprecated and will not receive more updates.
Please install fg-data-profiling via `pip install fg-data-profiling` and use `import data\_profiling` instead.

```
from ydata_profiling import ProfileReport
```

Section 2 – Project Input & Output

Input

Dataset

- Toronto Neighbourhood Profile 2016 (City of Toronto Open Data)
- File: `neighbourhood-profiles-2016-140-model.csv`

The dataset contains demographic, socioeconomic, education, housing, labour market, and population characteristics for all 140 Toronto neighbourhoods.

Output

This notebook will produce:

- A cleaned and structured machine-learning-ready dataset
- Selected socioeconomic indicators for predictive modelling
- Data quality assessment
- Feature selection justification
- Exploratory data analysis (EDA)
- YData Profiling report
- Exported dataset: `neighbourhood_profiles_2016_ml_ready.csv`

Data Loading

```
In [3]: file_path = "../data/raw/neighbourhood-profiles-2016-140-model.csv"

df_raw = pd.read_csv(file_path)

print("Dataset loaded successfully.")
print(f"Rows: {df_raw.shape[0]}")
print(f"Columns: {df_raw.shape[1]}")
```

Dataset loaded successfully.
Rows: 2383
Columns: 146

Standardize Column Names

The original Toronto Neighbourhood Profile dataset contains column names that differ slightly across dataset versions (*Characteristic* versus *Attribute*).

To maintain consistency throughout this project and ensure that the 2011 and 2016 notebooks follow the same structure, selected column names are standardized immediately after loading the dataset.

Standardizing column names improves code readability, simplifies feature extraction, and allows reusable code across multiple datasets.

```
In [4]: df_raw.rename(
        columns={
            "Characteristic": "Attribute"},
        inplace=True)
```

```
print("Column names standardized successfully.")
```

Column names standardized successfully.

Section 3 – Initial Data Inspection

Before performing data cleaning and feature selection, the raw dataset is examined to understand its overall structure, dimensions, data types, and content.

The objectives of this section are to:

- examine the size and structure of the dataset;
- inspect the first few records;
- identify variable names and data types;
- detect potential inconsistencies or formatting issues;
- establish a baseline understanding of the data before preprocessing.

This initial inspection provides the foundation for the subsequent data quality assessment and feature selection process.

```
In [5]: print("=" * 60)
print("Dataset Shape")
print("=" * 60)
print(df_raw.shape)

print("\n")

print("=" * 60)
print("First Five Rows")
print("=" * 60)
display(df_raw.head())

print("\n")

print("=" * 60)
print("Column Names")
print("=" * 60)
display(pd.DataFrame(df_raw.columns, columns=["Column Name"]))
```

```
=====
Dataset Shape
=====
(2383, 146)
```

```
=====
First Five Rows
=====
```

	_id	Category	Topic	Data Source	Attribute	City of Toronto	Agincourt North	Agincourt North
0	1	Neighbourhood Information	Neighbourhood Information	City of Toronto	Neighbourhood Number	NaN	129	
1	2	Neighbourhood Information	Neighbourhood Information	City of Toronto	TSNS2020 Designation	NaN	No Designation	Designation
2	3	Population	Population and dwellings	Census Profile 98-316-X2016001	Population, 2016	2,731,571	29,113	
3	4	Population	Population and dwellings	Census Profile 98-316-X2016001	Population, 2011	2,615,060	30,279	
4	5	Population	Population and dwellings	Census Profile 98-316-X2016001	Population Change 2011-2016	4.50%	-3.90%	

=====
Column Names
=====

Column Name	
0	_id
1	Category
2	Topic
3	Data Source
4	Attribute
...	...
141	Wychwood
142	Yonge-Eglinton
143	Yonge-St.Clair
144	York University Heights
145	Yorkdale-Glen Park

146 rows × 1 columns

Section 4 – Data Quality Assessment

Before selecting variables for predictive modelling, the quality of the dataset must be assessed to ensure that the data are complete, consistent, and suitable for analysis.

The objectives of this section are to:

- identify missing values;
- detect duplicate observations;
- examine data types;
- identify invalid or inconsistent values;
- evaluate the overall completeness of the dataset.

This assessment provides an initial understanding of the dataset's quality before more detailed profiling and feature selection are performed.

```
In [6]: print("=" * 60)
print("Dataset Information")
print("=" * 60)

df_raw.info()
```

```
=====
Dataset Information
=====
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2383 entries, 0 to 2382
Columns: 146 entries, _id to Yorkdale-Glen Park
dtypes: int64(1), object(145)
memory usage: 2.7+ MB
```

Missing Value Assessment

Missing values can reduce data quality and negatively affect exploratory analysis and predictive modelling.

This assessment identifies variables containing missing values and quantifies the proportion of missing observations within the dataset.

Understanding missingness at this stage helps determine whether variables should be retained, transformed, or excluded during feature selection.

```
In [7]: missing_values = (
        df_raw.isnull()
            .sum()
            .to_frame(name="Missing Values"))

missing_values["Missing (%)"] = (
    missing_values["Missing Values"] / len(df_raw) * 100).round(2)
```

```
missing_values = missing_values.sort_values(
    by="Missing Values",
    ascending=False)

display(missing_values.head(20))
```

	Missing Values	Missing (%)
Lansing-Westgate	56	2.35
Lawrence Park North	56	2.35
North Riverdale	56	2.35
North St. James Town	56	2.35
Oakridge	56	2.35
Oakwood Village	56	2.35
O'Connor-Parkview	56	2.35
Old East York	56	2.35
Palmerston-Little Italy	56	2.35
Parkwoods-Donalda	56	2.35
Pelmo Park-Humberlea	56	2.35
Playter Estates-Danforth	56	2.35
Pleasant View	56	2.35
Princess-Rosethorn	56	2.35
Regent Park	56	2.35
Rexdale-Kipling	56	2.35
Rockcliffe-Smythe	56	2.35
Niagara	56	2.35
Newtonbrook West	56	2.35
Newtonbrook East	56	2.35

```
In [8]: rows_with_missing = df_raw[df_raw.isnull().any(axis=1)]

print(f"Number of indicator rows containing missing values: {len(rows_with_missing)}")

display(
    rows_with_missing[
        ["Category", "Topic", "Attribute"]    ])
```

Number of indicator rows containing missing values: 62

	Category	Topic	Attribute
0	Neighbourhood Information	Neighbourhood Information	Neighbourhood Number
1	Neighbourhood Information	Neighbourhood Information	TSNS2020 Designation
946	Income	Income of individuals in 2015	Median total income in 2015 among recipie...
948	Income	Income of individuals in 2015	Median after-tax income in 2015 among reci...
951	Income	Income of individuals in 2015	Median market income in 2015 among recipie...
...	...	...	...
1960	Journey to work	Commuting destination	Total - Commuting destination for the employed...
1961	Journey to work	Commuting destination	Commute within census subdivision (CSD) of r...
1962	Journey to work	Commuting destination	Commute to a different census subdivision (C...
1963	Journey to work	Commuting destination	Commute to a different census subdivision (C...
1964	Journey to work	Commuting destination	Commute to a different province or territory

62 rows × 3 columns

Interpretation

A total of 62 indicator rows contain one or more missing values.

Most missing values occur in indicators that are not measured for every neighbourhood or represent aggregate statistics rather than neighbourhood-level observations.

The presence of missing values will be considered during feature selection, and only variables with sufficient neighbourhood-level coverage will be retained for predictive modelling.

Section 5 – Candidate Variable Screening

The objective of this section is to identify and screen candidate socioeconomic variables for inclusion in the machine learning dataset. Rather than selecting predictors solely through automated statistical procedures, a theory-informed screening strategy was adopted to

ensure that retained variables are relevant, consistently defined, and suitable for temporal analysis.

Candidate variables were identified from the Toronto Neighbourhood Profiles and evaluated according to predefined screening criteria before feature extraction. Statistical assessment of the retained variables is conducted separately in Notebook 2..

The Toronto Neighbourhood Profiles contain more than 2,300 indicator records, with each socioeconomic indicator stored as a separate row. To simplify variable identification and reduce manual searching, the `find_indicator()` helper function performs a keyword search within the **Attribute** column and returns all matching indicators. This allows candidate variables to be verified before they are extracted and transformed into the machine-learning-ready dataset.

```
In [9]: def find_indicator(keyword):
        """
        Search the Attribute column for a keyword.
        """
        results = df_raw[
            df_raw["Attribute"].str.contains(keyword, case=False, na=False) ]

        return results[["Category", "Topic", "Attribute"]]
```

5.1 Population

```
In [10]: find_indicator("Population, 2016")
```

```
Out[10]:
```

	Category	Topic	Attribute
2	Population	Population and dwellings	Population, 2016

5.2 Labour Force Indicators

```
In [11]: find_indicator("Participation rate")
```

```
Out[11]:
```

	Category	Topic	Attribute
1888	Labour	Labour force status	Participation rate
1896	Labour	Labour force status	Participation rate (Males)
1904	Labour	Labour force status	Participation rate (Females)

```
In [12]: find_indicator("Employment rate")
```

Out[12]:

	Category	Topic	Attribute
1889	Labour	Labour force status	Employment rate
1890	Labour	Labour force status	Unemployment rate
1897	Labour	Labour force status	Employment rate (Males)
1898	Labour	Labour force status	Unemployment rate (Males)
1905	Labour	Labour force status	Employment rate (Females)
1906	Labour	Labour force status	Unemployment rate (Females)

In [13]: find_indicator("Unemployment rate")

Out[13]:

	Category	Topic	Attribute
1890	Labour	Labour force status	Unemployment rate
1898	Labour	Labour force status	Unemployment rate (Males)
1906	Labour	Labour force status	Unemployment rate (Females)

5.3 Educational Attainment

In [14]: find_indicator("Bachelor")

Out[14]:

	Category	Topic	Attribute
1708	Education	Highest certificate, diploma or degree	University certificate or diploma below ba...
1709	Education	Highest certificate, diploma or degree	University certificate, diploma or degree ...
1710	Education	Highest certificate, diploma or degree	Bachelor's degree
1711	Education	Highest certificate, diploma or degree	University certificate or diploma above ...
1723	Education	Highest certificate, diploma or degree	University certificate or diploma below ba...
1724	Education	Highest certificate, diploma or degree	University certificate, diploma or degree ...
1725	Education	Highest certificate, diploma or degree	Bachelor's degree
1726	Education	Highest certificate, diploma or degree	University certificate or diploma above ...

In [15]: find_indicator("No certificate")

Out[15]:

	Category	Topic	Attribute
1703	Education	Highest certificate, diploma or degree	No certificate, diploma or degree
1715	Education	Highest certificate, diploma or degree	No certificate, diploma or degree

5.4 Immigration

In [16]: `find_indicator("Immigrants")`

Out[16]:

	Category	Topic	Attribute
1151	Immigration and citizenship	Immigrant status and period of immigration	Non-immigrants
1152	Immigration and citizenship	Immigrant status and period of immigration	Immigrants
1296	Immigration and citizenship	Admission category and applicant type	Economic immigrants
1299	Immigration and citizenship	Admission category and applicant type	Immigrants sponsored by family
1301	Immigration and citizenship	Admission category and applicant type	Other immigrants

5.5 Visible Minority

In [17]: `find_indicator("Visible")`

Out[17]:

	Category	Topic	Attribute
1335	Visible minority	Visible minority population	Total - Visible minority for the population in...
1336	Visible minority	Visible minority population	Total visible minority population
1346	Visible minority	Visible minority population	Visible minority; n.i.e.
1347	Visible minority	Visible minority population	Multiple visible minorities
1348	Visible minority	Visible minority population	Not a visible minority

5.6 Housing Indicators

In [18]: `find_indicator("Rent")`

Out[18]:

	Category	Topic	Attribute
94	Families, households and marital status	Family characteristics	Total lone-parent families by sex of parent
95	Families, households and marital status	Family characteristics	Female parent
96	Families, households and marital status	Family characteristics	Male parent
102	Families, households and marital status	Family characteristics	Lone-parent census families in private households
1081	Income	Income of economic families in 2015	Total - Income statistics in 2015 for lone-par...
1088	Income	Income of economic families in 2015	Median total income of lone-parent economic ...
1089	Income	Income of economic families in 2015	Median after-tax income of lone-parent econo...
1090	Income	Income of economic families in 2015	Average family size of lone-parent economic ...
1101	Income	Income of economic families in 2015	Total - Income statistics in 2015 for lone-par...
1102	Income	Income of economic families in 2015	Average total income of lone-parent economic...
1103	Income	Income of economic families in 2015	Average after-tax income of lone-parent econ...
1629	Housing	Household characteristics	Renter
1683	Education	Highest certificate, diploma or degree	Apprenticeship or trades certificate or di...
1693	Housing	Core housing need	Rate of unaffordable housing, renter-households
1705	Education	Highest certificate, diploma or degree	Trades certificate or diploma other than...
1706	Education	Highest certificate, diploma or degree	Certificate of Apprenticeship or Certifi...
1718	Education	Highest certificate, diploma or degree	Apprenticeship or trades certificate or di...
1719	Education	Highest certificate, diploma or degree	Trades certificate or diploma other than...
1721	Education	Highest certificate, diploma or degree	Certificate of Apprenticeship or Certifi...
1860	Education	Location of study compared with province or te...	Different than province or territory of ...

	Category	Topic	Attribute
1874	Education	Location of study compared with province or te...	Different than province or territory of ...
1945	Labour	Industry - North American Industry Classificat...	53 Real estate and rental and leasing
1962	Journey to work	Commuting destination	Commute to a different census subdivision (C...
1963	Journey to work	Commuting destination	Commute to a different census subdivision (C...
1964	Journey to work	Commuting destination	Commute to a different province or territory

In [19]: `find_indicator("Core housing")`

Out[19]:

	Category	Topic	Attribute
1681	Housing	Core housing need	Rate of core housing need
1682	Housing	Core housing need	Number of households in core housing need
1684	Housing	Core housing need	Core housing need, Unsuitable only
1685	Housing	Core housing need	Core housing need, Inadequate only
1686	Housing	Core housing need	Core housing need, Unaffordable only
1687	Housing	Core housing need	Core housing need, two or more dimensions

In [20]: `find_indicator("Unaffordable")`

Out[20]:

	Category	Topic	Attribute
1686	Housing	Core housing need	Core housing need, Unaffordable only
1688	Housing	Core housing need	Rate of unaffordable housing
1689	Housing	Core housing need	Unaffordable housing not in core need
1690	Housing	Core housing need	Unaffordable housing in core need
1691	Housing	Core housing need	Unaffordable housing not applicable for core need
1692	Housing	Core housing need	Rate of unaffordable housing, owner-households
1693	Housing	Core housing need	Rate of unaffordable housing, renter-households

5.7 Family Structure

In [21]: `find_indicator("lone-parent")`

Out[21]:

	Category	Topic	Attribute
94	Families, households and marital status	Family characteristics	Total lone-parent families by sex of parent
102	Families, households and marital status	Family characteristics	Lone-parent census families in private households
1081	Income	Income of economic families in 2015	Total - Income statistics in 2015 for lone-par...
1088	Income	Income of economic families in 2015	Median total income of lone-parent economic ...
1089	Income	Income of economic families in 2015	Median after-tax income of lone-parent econo...
1090	Income	Income of economic families in 2015	Average family size of lone-parent economic ...
1101	Income	Income of economic families in 2015	Total - Income statistics in 2015 for lone-par...
1102	Income	Income of economic families in 2015	Average total income of lone-parent economic...
1103	Income	Income of economic families in 2015	Average after-tax income of lone-parent econ...

5.8 Dependent Variable

In [22]: `find_indicator("low")`

Out[22]:

	Category	Topic	Attribute
1120	Income	Low income in 2015	Total - Low-income status in 2015 for the popu...
1125	Income	Low income in 2015	In low income based on the Low-income measure,...
1130	Income	Low income in 2015	Prevalence of low income based on the Low-inco...
1133	Income	Low income in 2015	In low income based on the Low-income cut-offs...
1139	Income	Low income in 2015	Prevalence of low income based on the Low-inco...
1708	Education	Highest certificate, diploma or degree	University certificate or diploma below ba...
1723	Education	Highest certificate, diploma or degree	University certificate or diploma below ba...
2304	Income	Income sources	Guaranteed Income Supplement (GIS) and S...
2305	Income	Income sources	Guaranteed Income Supplement (GIS) and S...
2307	Income	Income sources	Guaranteed Income Supplement (GIS) and S...

In [23]: `find_indicator("after-tax")`

Out[23]:

	Category	Topic	Attribute
947	Income	Income of individuals in 2015	Number of after-tax income recipients aged 1...
948	Income	Income of individuals in 2015	Median after-tax income in 2015 among reci...
960	Income	Income of individuals in 2015	Number of after-tax income recipients aged 1...
961	Income	Income of individuals in 2015	Average after-tax income in 2015 among rec...
984	Income	Income of individuals in 2015	Total - After-tax income groups in 2015 for th...
985	Income	Income of individuals in 2015	Without after-tax income
986	Income	Income of individuals in 2015	With after-tax income
987	Income	Income of individuals in 2015	Percentage with after-tax income
1019	Income	Income of households in 2015	Median after-tax income of households in 201...
1022	Income	Income of households in 2015	Median after-tax income of one-person ho...
1025	Income	Income of households in 2015	Median after-tax income of two-or-more-p...
1029	Income	Income of households in 2015	Average after-tax income of households in 20...
1032	Income	Income of households in 2015	Average after-tax income of one-person h...
1036	Income	Income of households in 2015	Average after-tax income of two-or-more...
1054	Income	Income of households in 2015	Total - Household after-tax income groups in 2...
1077	Income	Income of economic families in 2015	Median after-tax income of economic families...
1082	Income	Income of economic families in 2015	Median after-tax income of couple economic f...
1086	Income	Income of economic families in 2015	Median after-tax income of couple economic f...
1089	Income	Income of economic families in 2015	Median after-tax income of lone-parent econo...
1093	Income	Income of economic families in 2015	Average after-tax income of economic familie...
1097	Income	Income of economic families in 2015	Average after-tax income of couple economic ...

Category		Topic	Attribute
1100	Income	Income of economic families in 2015	Average after-tax income of couple economic ...
1103	Income	Income of economic families in 2015	Average after-tax income of lone-parent econ...
1106	Income	Income of economic families in 2015	Median after-tax income in 2015 for persons ...
2355	Income	Income sources	After-tax income: Population with an amount
2356	Income	Income sources	After-tax income: Average amount (\$)
2357	Income	Income sources	After-tax income: Aggregate amount (\$'000)

Interpretation

The candidate feature assessment confirmed that multiple neighbourhood-level socioeconomic indicators are available within the 2016 Toronto Neighbourhood Profile.

Each candidate variable was reviewed to verify its availability, completeness, and relevance for predictive modelling. Variables representing labour market conditions, educational attainment, immigration, visible minority population, housing conditions, family structure, and neighbourhood-level low-income prevalence were successfully identified.

The assessment also confirmed that some indicators available in earlier Census Profiles, such as median individual income and median after-tax income, are not consistently reported at the neighbourhood level in the 2016 dataset. Consequently, only variables with complete neighbourhood-level coverage will be retained for the final modelling dataset.

The verified indicators will now be organized into the final feature set for extraction and transformation.

Section 6 – Final Feature Selection

Following the candidate feature assessment, the final set of socioeconomic indicators is selected for predictive modelling.

Variables were retained based on four criteria:

- availability at the neighbourhood level;
- completeness and quality of the data;
- consistency with the 2011 Toronto Neighbourhood Profile where possible;
- relevance to neighbourhood socioeconomic vulnerability.

The selected variables consist of one target variable and multiple predictor variables that will be extracted and transformed into the final machine learning dataset.

```
In [24]: feature_mapping = {  
    "Population": 2,  
    "Participation_Rate": 1888,  
    "Employment_Rate": 1889,  
    "Unemployment_Rate": 1890,  
    "Bachelors_Degree": 1710,  
    "No_Diploma": 1704,  
    "Immigrant_Population": 1152,  
    "Visible_Minority": 1336,  
    "Renters": 1629,  
    "Core_Housing_Need": 1682,  
    "Unaffordable_Housing": 1688,  
    "Lone_Parent_Families": 102,  
    "Low_Income_AfterTax_Pct": 1130}
```

```
In [25]: feature_table = (pd.DataFrame({"Research Variable": feature_mapping.keys(), "Source  
display(feature_table)
```

	Research Variable	Source Row
0	Population	2
1	Participation_Rate	1888
2	Employment_Rate	1889
3	Unemployment_Rate	1890
4	Bachelors_Degree	1710
5	No_Diploma	1704
6	Immigrant_Population	1152
7	Visible_Minority	1336
8	Renters	1629
9	Core_Housing_Need	1682
10	Unaffordable_Housing	1688
11	Lone_Parent_Families	102
12	Low_Income_AfterTax_Pct	1130

Interpretation

A final set of 13 research variables was selected for predictive modelling, consisting of 12 independent variables and one dependent variable (**Low_Income_AfterTax_Pct**). Together with the Neighbourhood identifier, these variables form the machine learning-ready dataset used in the subsequent analyses.

The selected predictors capture labour market participation, income, educational attainment, immigration, visible minority status, housing conditions, family structure, and population characteristics.

These variables provide a comprehensive representation of neighbourhood vulnerability while maintaining compatibility with the 2011 Toronto Neighbourhood Profile for the predictive modelling stage.

The following section extracts the selected indicators and constructs the modelling dataset.

6.2 Selection of the Dependent Variable

The dependent variable selected for this study is **Low_Income_AfterTax_Pct (LIA-Tax%)**, representing the percentage of neighbourhood residents living below the **Statistics Canada Low-Income Measure, After-Tax (LIM-AT)**.

The selection of this variable was guided by both the study objectives and the social vulnerability literature. Cutter, Boruff, and Shirley (2003) describe social vulnerability as a multidimensional concept influenced by demographic, socioeconomic, educational, housing, labour market, and household characteristics. Within this framework, neighbourhood low-income prevalence represents an appropriate outcome because it reflects an important dimension of socioeconomic disadvantage that may be explained by these community characteristics.

LIA-Tax% is derived from the official Statistics Canada Low-Income Measure, After-Tax (LIM-AT), a nationally recognized indicator of relative low-income status. Furthermore, this variable is consistently available in both the 2011 and 2016 Toronto Neighbourhood Profiles datasets, enabling cross-sectional modelling and supporting the temporal validation conducted later in this study.

Accordingly, **Low_Income_AfterTax_Pct** is treated as the **dependent variable** and used as a **proxy indicator of neighbourhood socioeconomic vulnerability**, while recognizing that neighbourhood vulnerability is a broader multidimensional concept that cannot be fully represented by a single socioeconomic indicator.

References

- Cutter, S. L., Boruff, B. J., & Shirley, W. L. (2003). *Social vulnerability to environmental hazards*. *Social Science Quarterly*, 84(2), 242–261.
- Statistics Canada. (2016). *Census Dictionary – Low-Income Measure, After-Tax (LIM-AT)*.

Section 7 – Feature Extraction

Using the finalized feature mapping, the corresponding indicators are extracted from the 2016 Toronto Neighbourhood Profile.

Only the selected socioeconomic variables required for predictive modelling are retained. The resulting dataset will serve as the foundation for the machine learning-ready neighbourhood dataset.

7.1 Feature Extraction

```
In [26]: selected_df = df_raw.loc[
        list(feature_mapping.values())].copy()

print("=" * 60)
print("FEATURE EXTRACTION SUMMARY")
print("=" * 60)

print(f"\nSelected variables: {selected_df.shape[0]}")
print(f"Columns: {selected_df.shape[1]}")

display(selected_df[["_id", "Category", "Topic", "Attribute"]])
```

```
=====
FEATURE EXTRACTION SUMMARY
=====
```

```
Selected variables: 13
Columns: 146
```

	_id	Category	Topic	Attribute
2	3	Population	Population and dwellings	Population, 2016
1888	1889	Labour	Labour force status	Participation rate
1889	1890	Labour	Labour force status	Employment rate
1890	1891	Labour	Labour force status	Unemployment rate
1710	1711	Education	Highest certificate, diploma or degree	Bachelor's degree
1704	1705	Education	Highest certificate, diploma or degree	Secondary (high) school diploma or equivalen...
1152	1153	Immigration and citizenship	Immigrant status and period of immigration	Immigrants
1336	1337	Visible minority	Visible minority population	Total visible minority population
1629	1630	Housing	Household characteristics	Renter
1682	1683	Housing	Core housing need	Number of households in core housing need
1688	1689	Housing	Core housing need	Rate of unaffordable housing
102	103	Families, households and marital status	Family characteristics	Lone-parent census families in private households
1130	1131	Income	Low income in 2015	Prevalence of low income based on the Low-inco...

7.2 Remove Metadata

The selected indicators still contain descriptive metadata used for documentation purposes.

For machine learning, only the neighbourhood-level values are required. Therefore, the metadata columns are removed while preserving the selected socioeconomic indicators.

```
In [27]: model_data = selected_df.iloc[:, 4:].copy()
```

```
print("=" * 60)
print("MODEL DATA")
print("=" * 60)

print(model_data.shape)

display(model_data.head())
```

```
=====
MODEL DATA
=====
```

```
(13, 142)
```

	Attribute	City of Toronto	Agincourt North	Agincourt South-Malvern West	Alderwood	Annex	Banbury-Don Mills	Bathurst Manor
2	Population, 2016	2,731,571	29,113	23,757	12,054	30,526	27,695	15,873
1888	Participation rate	64.7	55.4	59	66.5	70.6	59.9	65
1889	Employment rate	59.3	50	53.2	62.4	65.8	55.6	60.3
1890	Unemployment rate	8.2	9.8	9.8	6.1	6.7	7.2	10.2
1710	Bachelor's degree	534610	4380	4210	1660	9135	6500	3075

```
In [28]: model_data = model_data.drop(columns=["Attribute"])
print(model_data.shape)
display(model_data.head())
```

(13, 141)

	City of Toronto	Agincourt North	Agincourt South-Malvern West	Alderwood	Annex	Banbury-Don Mills	Bathurst Manor	Bay Street Corridor
2	2,731,571	29,113	23,757	12,054	30,526	27,695	15,873	25,797
1888	64.7	55.4	59	66.5	70.6	59.9	65	62.7
1889	59.3	50	53.2	62.4	65.8	55.6	60.3	56.2
1890	8.2	9.8	9.8	6.1	6.7	7.2	7.2	10.2
1710	534610	4380	4210	1660	9135	6500	3075	8370

7.3 Rename Research Variables

The extracted indicators are relabelled using standardized research variable names.

Replacing the original dataset row identifiers with descriptive variable names improves readability and ensures consistency throughout the subsequent data preparation and predictive modelling stages.

```
In [29]: model_data.index = feature_mapping.keys()
```

```
print("=" * 60)
print("RESEARCH VARIABLE NAMES")
print("=" * 60)

display(model_data.head())
```

```
=====
RESEARCH VARIABLE NAMES
=====
```

	City of Toronto	Agincourt North	Agincourt South- Malvern West	Alderwood	Annex	Banbury- Don Mills	Bathur: Manc
Population	2,731,571	29,113	23,757	12,054	30,526	27,695	15,87
Participation_Rate	64.7	55.4	59	66.5	70.6	59.9	6
Employment_Rate	59.3	50	53.2	62.4	65.8	55.6	60.
Unemployment_Rate	8.2	9.8	9.8	6.1	6.7	7.2	7.
Bachelors_Degree	534610	4380	4210	1660	9135	6500	307

7.4 Transpose the Dataset

The selected indicators are currently stored as rows and neighbourhoods as columns.

To prepare the data for predictive modelling, the dataset is transposed so that each neighbourhood becomes one observation and each socioeconomic indicator becomes a feature.

This produces the standard machine learning format where:

- rows represent neighbourhoods;
- columns represent independent variables and the proxy dependent variable.

```
In [30]: ml_ready = model_data.T.copy()
```

```
print("=" * 60)
print("TRANPOSED DATASET")
print("=" * 60)

print(ml_ready.shape)

display(ml_ready.head())
```

=====

TRANPOSED DATASET

=====

(141, 13)

	Population	Participation_Rate	Employment_Rate	Unemployment_Rate	Bachelor
City of Toronto	2,731,571	64.7	59.3	8.2	
Agincourt North	29,113	55.4	50	9.8	
Agincourt South-Malvern West	23,757	59	53.2	9.8	
Alderwood	12,054	66.5	62.4	6.1	
Annex	30,526	70.6	65.8	6.7	

7.5 Create the Neighbourhood Identifier

Following the transposition, neighbourhood names are stored as the DataFrame index.

The index is converted into a dedicated column named **Neighbourhood**, allowing it to function as the unique identifier for each observation during subsequent analysis and modelling.

```
In [31]: ml_ready.reset_index(inplace=True)

ml_ready.rename(
    columns={"index": "Neighbourhood"},
    inplace=True)

print("=" * 60)
print("NEIGHBOURHOOD IDENTIFIER CREATED")
print("=" * 60)

display(ml_ready.head())
```

=====

NEIGHBOURHOOD IDENTIFIER CREATED

=====

	Neighbourhood	Population	Participation_Rate	Employment_Rate	Unemployment_Rate	
0	City of Toronto	2,731,571	64.7	59.3	8.2	
1	Agincourt North	29,113	55.4	50	9.8	
2	Agincourt South-Malvern West	23,757	59	53.2	9.8	
3	Alderwood	12,054	66.5	62.4	6.1	
4	Annex	30,526	70.6	65.8	6.7	

7.6 Convert Variables to Numeric

Neighbourhood identifiers remain as text, while all socioeconomic indicators are converted to numeric values.

This ensures compatibility with statistical analyses, machine learning algorithms, and data visualization techniques.

```
In [32]: for column in ml_ready.columns[1:]:
          ml_ready[column] = (
              ml_ready[column]
              .astype(str)
              .str.replace(",", "", regex=False)
          )

          ml_ready[column] = pd.to_numeric(
              ml_ready[column],
              errors="coerce"
          )

print("=" * 60)
print("DATA TYPES")
print("=" * 60)

display(ml_ready.dtypes)
```

```
=====
DATA TYPES
=====
```

```

Neighbourhood      object
Population         int64
Participation_Rate float64
Employment_Rate    float64
Unemployment_Rate  float64
Bachelors_Degree   int64
No_Diploma          int64
Immigrant_Population int64
Visible_Minority    int64
Renters            int64
Core_Housing_Need  int64
Unaffordable_Housing float64
Lone_Parent_Families int64
Low_Income_AfterTax_Pct float64
dtype: object

```

7.7 Data Quality Assessment

Prior to exporting the modelling dataset, several quality assurance checks are performed.

The dataset is assessed for:

- missing values;
- duplicate neighbourhood records;
- overall dataset structure.

These validation steps ensure that the machine learning dataset is complete and suitable for predictive modelling.

```

In [33]: print("=" * 60)
          print("FINAL DATASET SHAPE")
          print("=" * 60)

          print(ml_ready.shape)

```

```

=====
FINAL DATASET SHAPE
=====
(141, 14)

```

7.8 Remove the City of Toronto Aggregate

Remove the City of Toronto aggregate observation before exporting the machine learning-ready dataset. The aggregate row summarizes city-wide socioeconomic conditions rather than representing an individual neighbourhood and is therefore excluded to ensure that all subsequent analyses are conducted exclusively at the neighbourhood level.

```

In [34]: # Remove the City of Toronto aggregate row

ml_ready = (ml_ready[ml_ready["Neighbourhood"] != "City of Toronto"].reset_index(drop=True))

print("=" * 60)

```

```
print("CITY OF TORONTO AGGREGATE REMOVED")
print("=" * 60)

print(f"Rows: {ml_ready.shape[0]}")
print(f"Columns: {ml_ready.shape[1]}")

display(ml_ready.head())
```

```
=====
CITY OF TORONTO AGGREGATE REMOVED
=====
```

Rows: 140
Columns: 14

	Neighbourhood	Population	Participation_Rate	Employment_Rate	Unemployment_Rate	I
0	Agincourt North	29113	55.4	50.0	9.8	
1	Agincourt South-Malvern West	23757	59.0	53.2	9.8	
2	Alderwood	12054	66.5	62.4	6.1	
3	Annex	30526	70.6	65.8	6.7	
4	Banbury-Don Mills	27695	59.9	55.6	7.2	

7.9 Descriptive Statistics for selected socioeconomic variables

The following exploratory analysis provides a statistical summary and visual examination of the cleaned dataset prior to predictive modelling.

```
In [35]: # Descriptive Statistics
descriptive_stats = ml_ready.describe().T.round(2)

print(descriptive_stats)

descriptive_stats
```

	count	mean	std	min	25%	\
Population	140.0	19511.22	10033.59	6577.0	12019.50	
Participation_Rate	140.0	64.55	6.11	52.2	59.88	
Employment_Rate	140.0	59.27	6.52	47.3	54.68	
Unemployment_Rate	140.0	8.30	1.90	4.5	6.90	
Bachelors_Degree	140.0	3819.00	3046.74	445.0	2053.75	
No_Diploma	140.0	4007.61	2281.24	1190.0	2372.50	
Immigrant_Population	140.0	9042.89	6141.42	1795.0	4621.25	
Visible_Minority	140.0	9899.04	8355.39	1100.0	3633.75	
Renters	140.0	3755.68	3008.98	280.0	1946.25	
Core_Housing_Need	140.0	1720.14	1002.88	115.0	967.50	
Unaffordable_Housing	140.0	35.07	6.94	17.5	31.08	
Lone_Parent_Families	140.0	1090.39	660.31	220.0	640.00	
Low_Income_AfterTax_Pct	140.0	19.51	7.89	4.5	14.10	

	50%	75%	max
Population	16749.50	23854.50	65913.0
Participation_Rate	63.60	69.12	86.6
Employment_Rate	58.55	64.30	82.7
Unemployment_Rate	8.20	9.62	14.6
Bachelors_Degree	3257.50	4385.00	27280.0
No_Diploma	3330.00	5197.50	13170.0
Immigrant_Population	7502.50	11327.50	31040.0
Visible_Minority	7122.50	12505.00	40510.0
Renters	3232.50	4338.75	23930.0
Core_Housing_Need	1432.50	2287.50	5335.0
Unaffordable_Housing	35.60	38.52	56.5
Lone_Parent_Families	870.00	1365.00	3460.0
Low_Income_AfterTax_Pct	18.55	23.95	45.5

Out[35]:

	count	mean	std	min	25%	50%	75%
Population	140.0	19511.22	10033.59	6577.0	12019.50	16749.50	23854.50
Participation_Rate	140.0	64.55	6.11	52.2	59.88	63.60	69.12
Employment_Rate	140.0	59.27	6.52	47.3	54.68	58.55	64.30
Unemployment_Rate	140.0	8.30	1.90	4.5	6.90	8.20	9.62
Bachelors_Degree	140.0	3819.00	3046.74	445.0	2053.75	3257.50	4385.00
No_Diploma	140.0	4007.61	2281.24	1190.0	2372.50	3330.00	5197.50
Immigrant_Population	140.0	9042.89	6141.42	1795.0	4621.25	7502.50	11327.50
Visible_Minority	140.0	9899.04	8355.39	1100.0	3633.75	7122.50	12505.00
Renters	140.0	3755.68	3008.98	280.0	1946.25	3232.50	4338.75
Core_Housing_Need	140.0	1720.14	1002.88	115.0	967.50	1432.50	2287.50
Unaffordable_Housing	140.0	35.07	6.94	17.5	31.08	35.60	38.52
Lone_Parent_Families	140.0	1090.39	660.31	220.0	640.00	870.00	1365.00
Low_Income_AfterTax_Pct	140.0	19.51	7.89	4.5	14.10	18.55	23.95



```
In [36]: descriptive_stats = ml_ready.describe().T.round(2)
descriptive_stats.index.name = "Variable"
descriptive_stats
```

Out[36]:

	count	mean	std	min	25%	50%	75%
Variable							
Population	140.0	19511.22	10033.59	6577.0	12019.50	16749.50	23854.50
Participation_Rate	140.0	64.55	6.11	52.2	59.88	63.60	69.12
Employment_Rate	140.0	59.27	6.52	47.3	54.68	58.55	64.30
Unemployment_Rate	140.0	8.30	1.90	4.5	6.90	8.20	9.62
Bachelors_Degree	140.0	3819.00	3046.74	445.0	2053.75	3257.50	4385.00
No_Diploma	140.0	4007.61	2281.24	1190.0	2372.50	3330.00	5197.50
Immigrant_Population	140.0	9042.89	6141.42	1795.0	4621.25	7502.50	11327.50
Visible_Minority	140.0	9899.04	8355.39	1100.0	3633.75	7122.50	12505.00
Renters	140.0	3755.68	3008.98	280.0	1946.25	3232.50	4338.75
Core_Housing_Need	140.0	1720.14	1002.88	115.0	967.50	1432.50	2287.50
Unaffordable_Housing	140.0	35.07	6.94	17.5	31.08	35.60	38.52
Lone_Parent_Families	140.0	1090.39	660.31	220.0	640.00	870.00	1365.00
Low_Income_AfterTax_Pct	140.0	19.51	7.89	4.5	14.10	18.55	23.95

the table summarizes the descriptive statistics for the 13 socioeconomic indicators across the 140 Toronto neighbourhoods included in the analysis. The dataset contains complete observations for every variable, with no missing values remaining after the data cleaning and preprocessing stages. This confirms that the final analytical dataset was suitable for subsequent statistical and machine learning analyses.

The descriptive statistics demonstrate considerable variability across neighbourhoods, reflecting substantial socioeconomic diversity within the City of Toronto. For example, Low Income After-Tax Percentage, the study's target variable, ranged from 4.5% to 45.5%, with a mean of 19.51% and a standard deviation of 7.89 percentage points. This wide range indicates that while some neighbourhoods experience relatively low levels of economic disadvantage, others face significantly higher rates of low-income prevalence.

Several predictor variables also exhibited substantial variation. Population ranged from 6,577 to 65,913 residents, while Immigrant Population, Visible Minority, and Renters showed large standard deviations, suggesting considerable demographic differences among neighbourhoods. Housing-related indicators, including Core Housing Need and Unaffordable Housing, likewise varied substantially, indicating unequal housing conditions

across the city. In contrast, labour market indicators such as Participation Rate, Employment Rate, and Unemployment Rate displayed more moderate variability, suggesting relatively consistent labour force characteristics across many neighbourhoods.

Overall, the descriptive statistics confirm that the dataset contains sufficient variation across demographic, socioeconomic, labour market, educational, and housing characteristics to support meaningful exploratory analysis, dimensionality reduction, clustering, and predictive modelling. The observed variability also supports the use of neighbourhood-level machine learning techniques to investigate socioeconomic vulnerability across Toronto

7.10. Distribution of Low Income After-Tax Percentage

```
In [37]: %matplotlib inline
import matplotlib.pyplot as plt

mean_lim = ml_ready["Low_Income_AfterTax_Pct"].mean()

fig, ax = plt.subplots(figsize=(8,5))

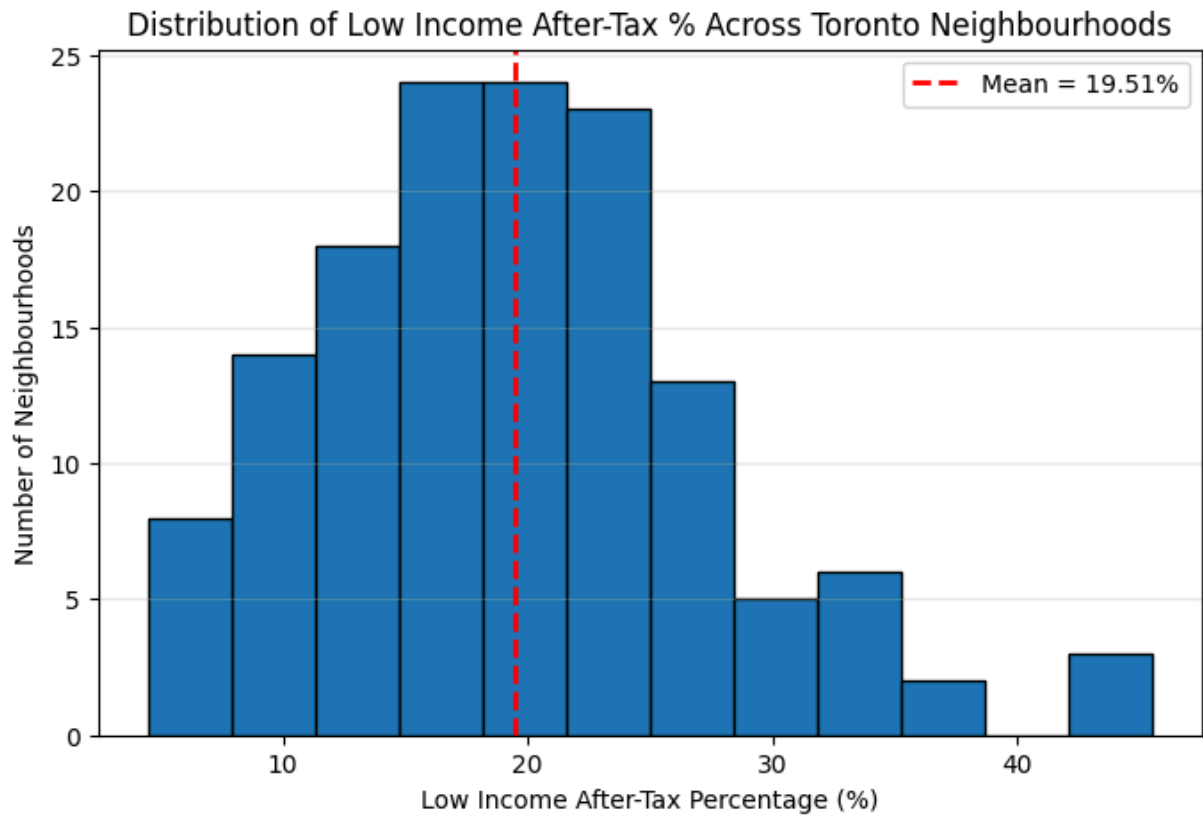
ax.hist(
    ml_ready["Low_Income_AfterTax_Pct"],
    bins=12,
    edgecolor="black")

ax.axvline(
    mean_lim,
    color="red",
    linestyle="--",
    linewidth=2,
    label=f"Mean = {mean_lim:.2f}%")

ax.set_title("Distribution of Low Income After-Tax % Across Toronto Neighbourhoods")
ax.set_xlabel("Low Income After-Tax Percentage (%)")
ax.set_ylabel("Number of Neighbourhoods")

ax.legend()
ax.grid(axis="y", alpha=0.3)

plt.show()
```



the histogram illustrates the distribution of Low Income After-Tax Percentage (LIM-AT) across the 140 Toronto neighbourhoods included in the analysis. Consistent with the descriptive statistics presented in Table 6.1, most neighbourhoods are concentrated between approximately 15% and 25%, with the average low-income rate of 19.51% indicated by the red dashed line.

The distribution exhibits a slight right skew, with a small number of neighbourhoods experiencing substantially higher low-income rates than the majority. This suggests that while most Toronto neighbourhoods have moderate levels of low-income prevalence, a limited number face considerably greater socioeconomic challenges. The observed variation confirms that the response variable contains sufficient diversity to support subsequent Principal Component Analysis, cluster analysis, and predictive modelling.

8. Export Machine Learning Dataset

Following data preparation, feature extraction, restructuring, and validation, the finalized machine learning-ready dataset is exported for use in the subsequent stages of the capstone project.

The exported dataset serves as the standardized analytical input for all remaining notebooks, ensuring that feature relationship analysis, dimensionality reduction, clustering, predictive modelling, and temporal validation are performed using a consistent and reproducible neighbourhood-level dataset.

The exported dataset contains:

- **140 neighbourhood observations** (City of Toronto aggregate removed);
- **12 independent variables** representing demographic, socioeconomic, labour market, housing, education, immigration, and ethnocultural characteristics;
- **1 dependent variable** representing neighbourhood-level low-income prevalence (Low-Income After-Tax Percentage, LIA-Tax%), used as a **proxy indicator of neighbourhood socioeconomic vulnerability**;
- **1 identifier column** (Neighbourhood).

Independent Variables (X):

- Population
- Participation_Rate
- Employment_Rate
- Unemployment_Rate
- Bachelors_Degree
- No_Diploma
- Visible_Minority
- Renters
- Immigrant_Population
- Lone_Parent_Families
- Core_Housing_Need
- Unaffordable_Housing

Dependent Variable (Y):

- Low_Income_AfterTax_Pct

Exporting the dataset at this stage establishes a single, reproducible source of truth for all subsequent analyses and ensures methodological consistency throughout the capstone project.

```
In [38]: output_path = "../data/processed/neighbourhood_profiles_2016_ml_ready.csv"

ml_ready.to_csv(
    output_path,
    index=False)

print("=" * 60)
print("EXPORT COMPLETE")
print("=" * 60)

print(f"Dataset saved to:\n{output_path}")

print("\nDataset Shape:")
print(ml_ready.shape)
```

```
display(ml_ready.head())
```

```
=====
EXPORT COMPLETE
=====
```

```
Dataset saved to:
```

```
../..data/processed/neighbourhood_profiles_2016_ml_ready.csv
```

```
Dataset Shape:
```

```
(140, 14)
```

	Neighbourhood	Population	Participation_Rate	Employment_Rate	Unemployment_Rate	I
0	Agincourt North	29113	55.4	50.0	9.8	
1	Agincourt South-Malvern West	23757	59.0	53.2	9.8	
2	Alderwood	12054	66.5	62.4	6.1	
3	Annex	30526	70.6	65.8	6.7	
4	Banbury-Don Mills	27695	59.9	55.6	7.2	

9. Automated Data Profiling

Following the export of the finalized machine learning-ready dataset, an automated exploratory data analysis (EDA) is performed using **YData Profiling** to independently validate the exported analytical dataset.

The original Toronto Neighbourhood Profiles dataset was not suitable for automated profiling because it was organized with socioeconomic indicators as rows and neighbourhoods as columns, rather than following the conventional observation-by-variable structure expected by automated profiling tools. Consequently, manual exploratory data analysis was performed during the data preparation stage to understand the dataset structure, identify the required socioeconomic indicators, assess data quality, and prepare the dataset for transformation.

Once the dataset had been restructured into a machine learning-ready format, automated profiling became appropriate. Unlike the earlier manual exploratory analysis, **YData Profiling** provides a comprehensive assessment of the finalized dataset by generating descriptive statistics, variable distributions, correlation analysis, missing value diagnostics, duplicate detection, and data quality warnings.

This automated profiling step serves as an additional quality assurance procedure to verify that the exported dataset is complete, internally consistent, and suitable for subsequent statistical analyses and predictive modelling.

The generated HTML report is stored in the project's **reports** directory and serves as supplementary documentation supporting the reproducibility and quality assurance of this study.

```
In [39]: from ydata_profiling import ProfileReport
```

```
In [40]: validation_df = pd.read_csv(output_path)

profile = ProfileReport(
    validation_df,
    title="Toronto Neighbourhood Profile (2016) - Machine Learning Dataset",
    explorative=True)

print("=" * 60)
print("YDATA PROFILING")
print("=" * 60)
print("Profiling report generated successfully.")
```

```
=====
YDATA PROFILING
=====
Profiling report generated successfully.
```

```
In [41]: report_path = "../reports/Toronto_ML_Profile_Report_2016.html"

profile.to_file(report_path)

print("=" * 60)
print("REPORT SAVED")
print("=" * 60)
print(f"Location:\n{report_path}")
```

```
Summarize dataset:  0%|          | 0/5 [00:00<?, ?it/s]
100%|██████████| 14/14 [00:00<00:00, 41.42it/s]
Generate report structure:  0%|          | 0/1 [00:00<?, ?it/s]
Render HTML:  0%|          | 0/1 [00:00<?, ?it/s]
Export report to file:  0%|          | 0/1 [00:00<?, ?it/s]
=====
REPORT SAVED
=====
Location:
../reports/Toronto_ML_Profile_Report_2016.html
```

Interpretation

The automated YData Profiling report successfully validated the exported machine learning-ready dataset. Unlike the original Toronto Neighbourhood Profiles dataset, which required manual exploratory analysis because of its indicator-oriented structure, the transformed dataset follows the conventional observation-by-variable format required for automated profiling.

The profiling report confirmed the dataset structure, variable distributions, descriptive statistics, missing values, duplicate observations, and correlation patterns. These findings verified that the data transformation process was successful and that the exported dataset is internally consistent and suitable for neighbourhood-level statistical analyses and predictive modelling.

The generated HTML report serves as supplementary quality assurance documentation, complementing the earlier manual exploratory data analysis and providing independent validation of the finalized analytical dataset.

10. Conclusion

Summary

This notebook completed the data preparation and exploratory data analysis (EDA) phase of the project by transforming the raw 2016 Toronto Neighbourhood Profiles dataset into a structured machine learning-ready dataset suitable for neighbourhood-level statistical analysis and predictive modelling.

The original dataset required substantial preprocessing because it was organized with socioeconomic indicators as rows and neighbourhoods as columns rather than in the conventional observation-by-variable format. Consequently, manual exploratory data analysis was performed to understand the dataset structure, identify the required socioeconomic indicators, assess data quality, and prepare the data for transformation into a machine learning-ready format.

Following feature extraction and data restructuring, the dataset was transformed into a conventional analytical format consisting of neighbourhoods as observations and socioeconomic indicators as variables. Data quality was assessed through both manual validation procedures and an automated YData Profiling report, confirming that the finalized dataset contained the expected variable structure, appropriate data types, and no missing values within the selected variables.

The finalized machine learning-ready dataset consists of:

- **140 neighbourhood observations** (City of Toronto aggregate removed);
- **12 independent variables** representing demographic, socioeconomic, labour market, housing, education, immigration, and ethnocultural characteristics;
- **1 dependent variable, Low_Income_AfterTax_Pct**, representing neighbourhood-level low-income prevalence and used in this study as a **proxy indicator of neighbourhood socioeconomic vulnerability**;
- **1 identifier column** (Neighbourhood).

The finalized dataset has been exported as the standardized analytical input for all subsequent notebooks, ensuring a consistent, reproducible, and neighbourhood-level

analytical workflow throughout the remainder of the capstone project.

Project Analytical Workflow

Raw Toronto Neighbourhood Profiles Dataset (2016)



Manual Exploratory Data Analysis
(CIND110; CIND123)



Feature Extraction & Variable Selection
(CMTH 642)



Data Cleaning, Validation & Transformation
(CIND110)



Machine Learning-Ready Dataset
(140 Neighbourhoods)



Automated YData Profiling
(Clemente et al., 2023)



Notebook 2
Feature Relationship Analysis
(CIND123)



Notebook 3
PCA & Neighbourhood Vulnerability Assessment
(CMTH 642 Jolliffe & Cadima, 2016;
MacQueen, 1967;
Everitt et al., 2011)



Notebook 4
Predictive Modelling
(Linear Regression & Random Forest)



Notebook 5
2011 Machine Learning Dataset Preparation



Notebook 6
Temporal Validation & Comparative Analysis





Notebook 7

Temporal Predictive Model Validation